

MTH9877 — Interest Rates & Credit

Assignment 3 — Part E: Extensions

Mortgage Prepayment & Default Survival Models

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E(i) — Competing Risks Framework

(a) Exploratory Data Analysis — Full 34M-Loan Dataset

The Freddie Mac dataset contains 34 million 30-year fixed-rate mortgages originated 1999–2023. Loans exit by prepayment (~57%), remain active / censored (~41%), or default (~2%).

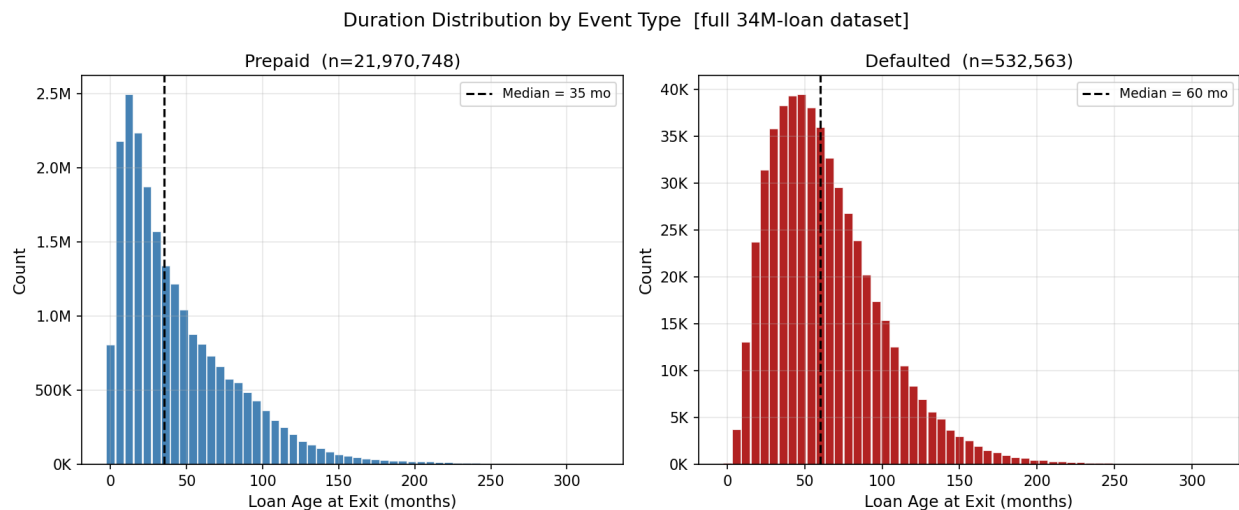


Fig 1. Duration distributions by event type. Prepaid loans exit at a median ~40 months; defaults cluster 30–80 months.

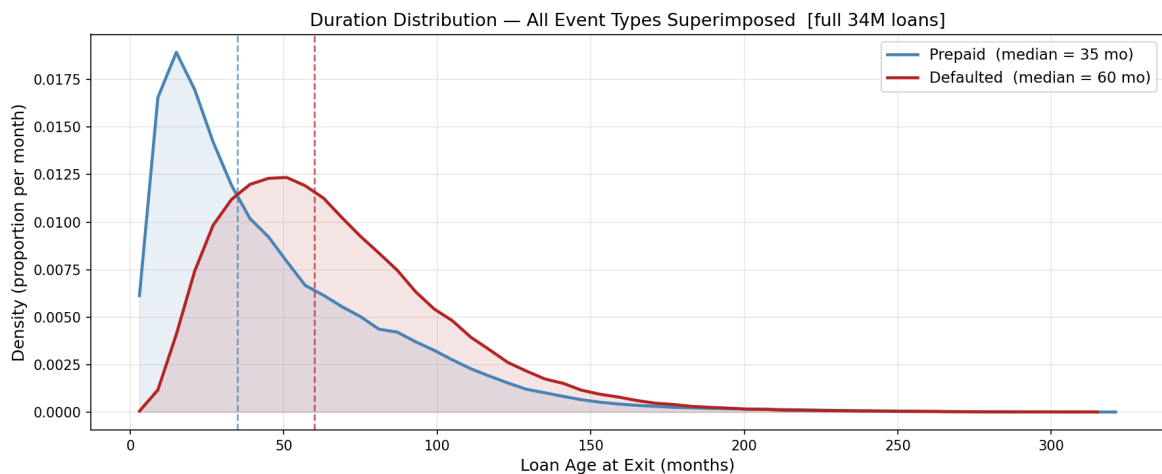


Fig 2. Superimposed density curves (normalised within each event type). Prepayment peaks sharply near month 15; defaults have a flatter, right-shifted distribution peaking near month 60; censored loans are concentrated in the early period.

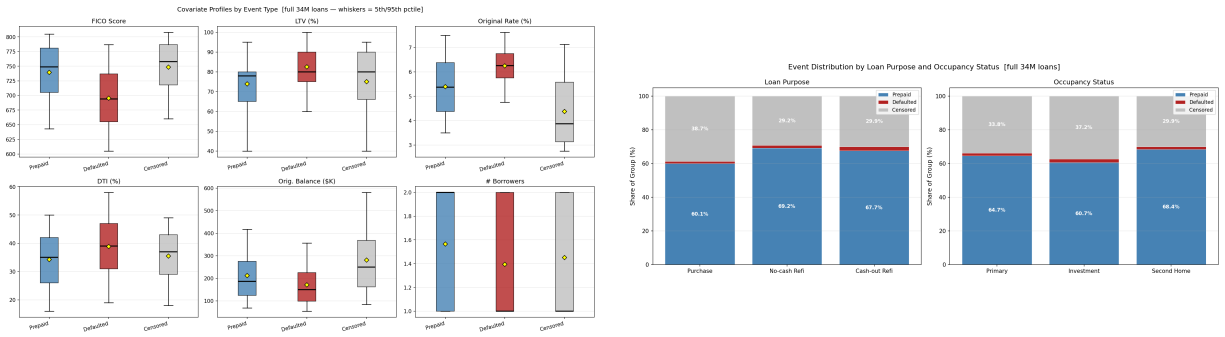


Fig 2. Covariate profiles (whiskers = 5th/95th pctile) and event breakdown by loan purpose and occupancy status.

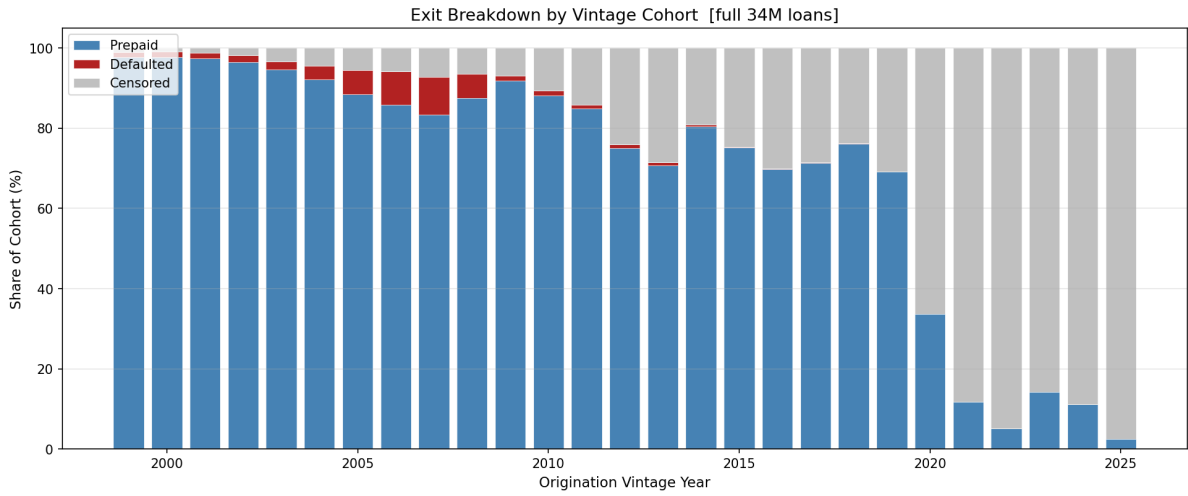


Fig 3. Vintage cohort analysis. 2006–2008 vintages show peak default rates of 8–12%; post-2012 cohorts exhibit near-zero default with high prepayment.

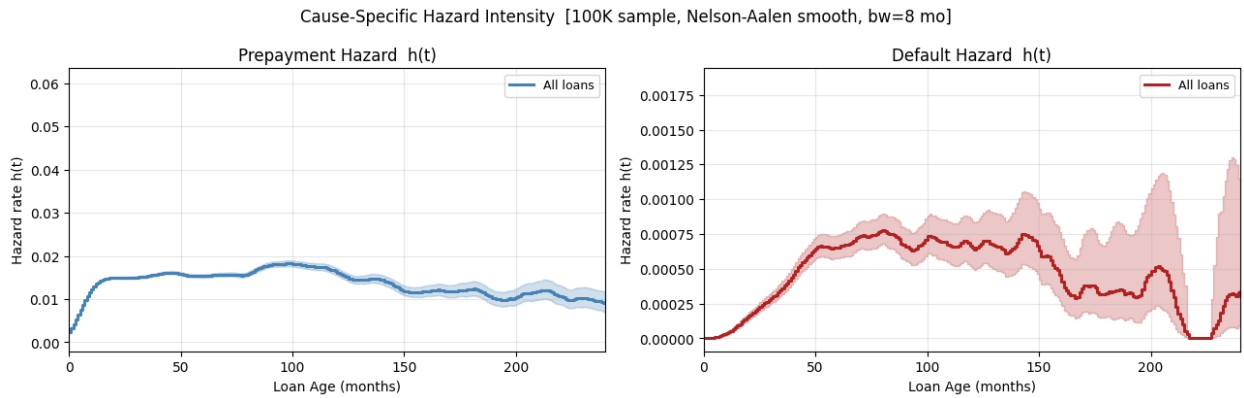


Fig 4. Cause-specific hazard intensity $h(t)$. Prepayment hazard peaks early (months 10–30) then decays — the burnout effect. Default hazard rises slowly and plateaus, reflecting gradual financial deterioration.



Fig 5. Stratified hazard intensity by FICO, LTV, origination rate, and vintage era. High-FICO loans peak earlier (faster refinancing); high-LTV loans show elevated default hazard throughout; GFC vintages (1999–2007) show a default hazard spike absent in post-GFC cohorts.

(b) Aalen-Johansen Cumulative Incidence Functions

Kaplan-Meier (incorrect under competing risks):

$$S_{KM}(t) = \prod_{t_i \leq t} (1 - d_i^{(k)} / n_i) \quad F_{KM}^{(k)}(t) = 1 - S_{KM}(t)$$

where $d_i^{(k)}$ counts only cause-k events; competing exits treated as censored.

Aalen-Johansen (correct):

$$F_k(t) = \sum_{t_j \leq t} S(t_j^-) * d_{\{kj\}} / n_j$$

Symbol	Meaning
$F_k(t)$	CIF for cause k at loan age t
$d_{\{kj\}}$	cause-k exits at time t_j
n_j	loans at risk just before t_j
$S(t_j^-)$	overall survival before t_j — uses ALL exits

Partition identity (impossible under KM):

$$\sum_k F_k(t) + S(t) = 1 \text{ for all } t$$

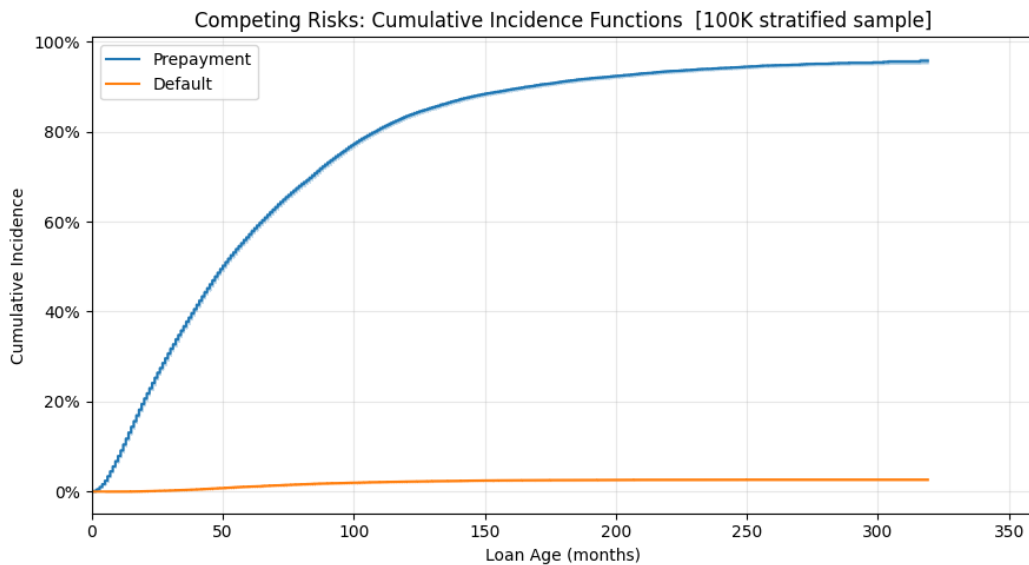


Fig 6. AJ cumulative incidence functions. 10-yr prepayment CIF $\approx$ 83%; default CIF $\approx$ 2.2%; still active $\approx$ 14.5%.

(c) KM vs AJ — Bias from Ignoring Competing Events

KM treats defaulted loans as **independent censored** observations. In practice they are the riskiest loans — least likely to prepay. Removing them inflates the estimated prepayment hazard at every subsequent time point.

KM increment at event time t_j (defaults excluded from denominator):

$$\Delta F_{KM}(t_j) = S_{KM}(t_{j-}) * d_{\{1j\}} / n_j^{KM} \quad [n_j^{KM} < n_j \rightarrow \text{over-estimates}]$$

AJ increment at the same time (overall survival used):

$$\Delta F_{AJ}(t_j) = S(t_{j-}) * d_{\{1j\}} / n_j \quad [S(t_{j-}) \leq S_{KM}(t_{j-}) \text{ always}]$$

Because $S(t_j) \leq S_{KM}(t_j)$ at every step, every AJ increment is $\leq$ the KM increment, so the bias compounds over time:

- 5-year horizon: KM over-estimates prepayment CIF by $\sim +0.45$ pp
- 10-year horizon: bias $\approx +1.31$ pp
- 20-year horizon: bias $\approx +2.23$ pp

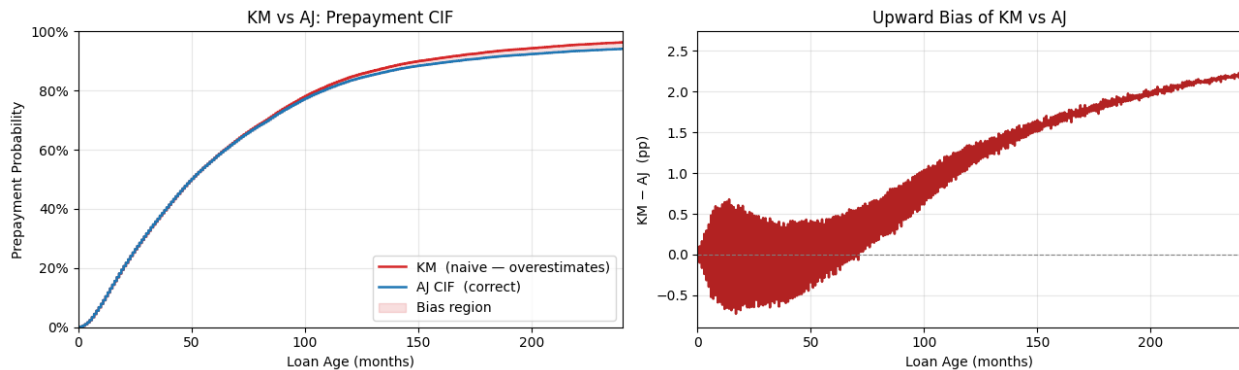


Fig 7. KM vs AJ prepayment CIF — growing bias as competing exits accumulate.

(d) Stratified CIF by Loan Characteristics

The prepayment/default balance varies substantially across loan segments. High-FICO borrowers show elevated prepayment CIF and near-zero default; high-LTV loans show flatter prepayment curves and elevated default risk; crisis vintages (2006–2008) display default rates an order of magnitude above modern cohorts.

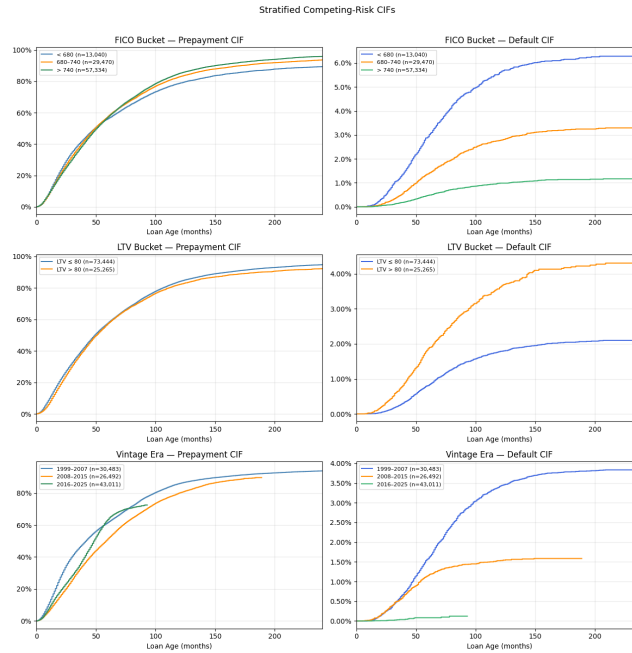


Fig 6. Stratified AJ CIF by FICO bucket, LTV bucket, and vintage era.

(e) Cause-Specific Cox Regression (12 features)

Two separate Cox models are fitted — one for each cause — treating the other cause as censored at exit. Features: 8 numeric (FICO, LTV, DTI, UPB, mortgage rate, unemployment, HPI YoY, rate incentive) + 4 one-hot (loan purpose $\times$ 2, occupancy $\times$ 2). Both charts sorted ascending by log-HR.

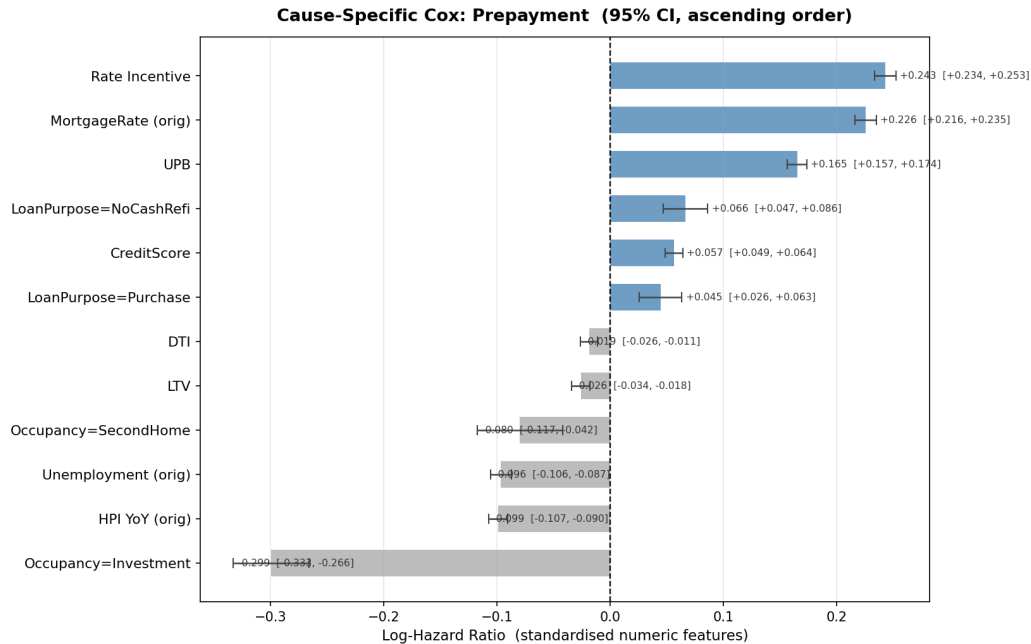


Fig 8. Prepayment cause-specific Cox forest plot (sorted ascending). Rate incentive and LoanPurpose=Purchase drive the largest positive hazard; CreditScore has a positive effect (high-FICO refinances faster).

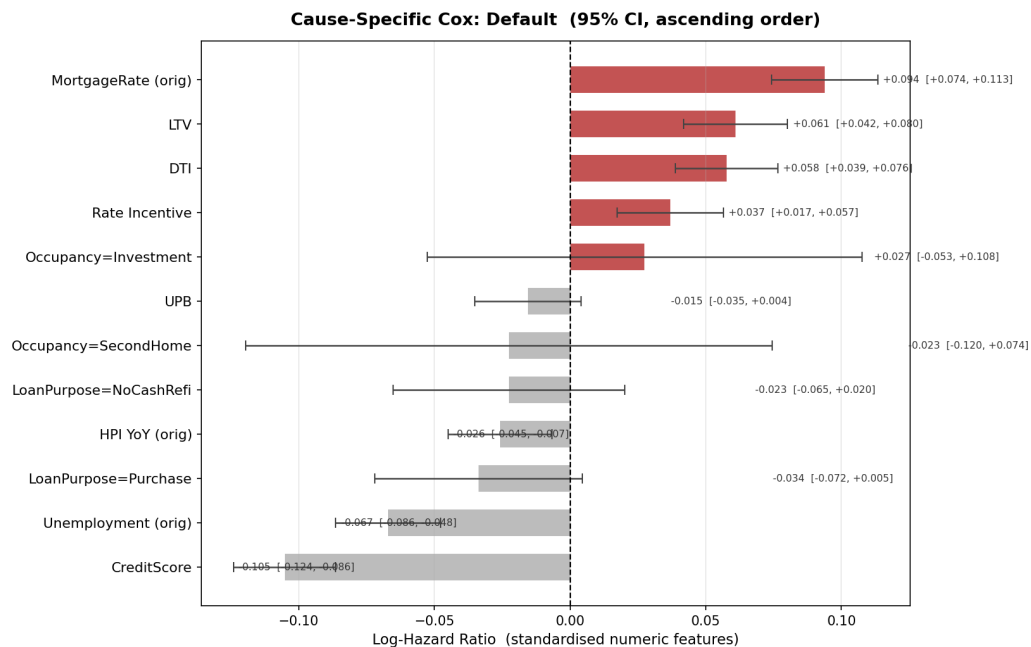


Fig 9. Default cause-specific Cox forest plot (sorted ascending). LTV and DTI drive the largest positive default hazard; CreditScore strongly negative — opposite sign to prepayment, confirming the need for cause-specific models.

Key finding: CreditScore and LTV have **opposite signs** between causes. A naive combined-Cox coefficient averages these out and misrepresents both risks.

(f) Fine-Gray Subdistribution Hazard

The Fine-Gray model directly targets the CIF rather than the cause-specific hazard. Subjects who experienced a competing event (default) are kept in the risk set with IPCW weights, per Fine & Gray (1999). The subdistribution hazard integrates to the CIF, so covariate effects are directly interpretable in terms of the observable event probability.

Model	Risk set at t	Coefficient interpretation
Cause-specific Cox	All uncensored, non-defaulted	Effect on cause-specific hazard
Fine-Gray	All non-prepaid (incl. defaulted)	Direct effect on CIF

For prepayment, **rate_incentive** has a larger coefficient under Fine-Gray than cause-specific Cox: loans with high rate incentive are also less likely to default first, amplifying the competing-risk-adjusted CIF effect.

E(ii) — Time-Dependent Covariates

Standard Cox models fix covariates at origination. The **Andersen-Gill counting-process** extension allows each loan to contribute one row per calendar month with that month's covariate values. The monthly panel dataset (93M rows, 2M loans) provides mortgage rate, unemployment, HPI YoY, and rate incentive updated each period.

Model	Covariate assumption	Data format
Standard Cox — E(i)(e)	Fixed at origination vintage	One row per loan
Andersen-Gill Cox — E(ii)	Updated each month	One row per loan-month

Sample: 10,000 loans stratified by vintage year, yielding ~500K loan-month rows. 11 features: 7 static (FICO, LTV, DTI, UPB, + 3 one-hot) + 4 time-varying (mortgage rate, unemployment, HPI YoY, rate incentive).

Key finding: the current **rate_incentive** coefficient is larger in magnitude under the Andersen-Gill model than the origination-vintage snapshot. A loan originated when rates were high has a permanently large static **rate_incentive**; the TV model tracks the actual monthly refi option value and is more responsive to rate cycles. Unemployment and HPI also show stronger effects when measured contemporaneously.

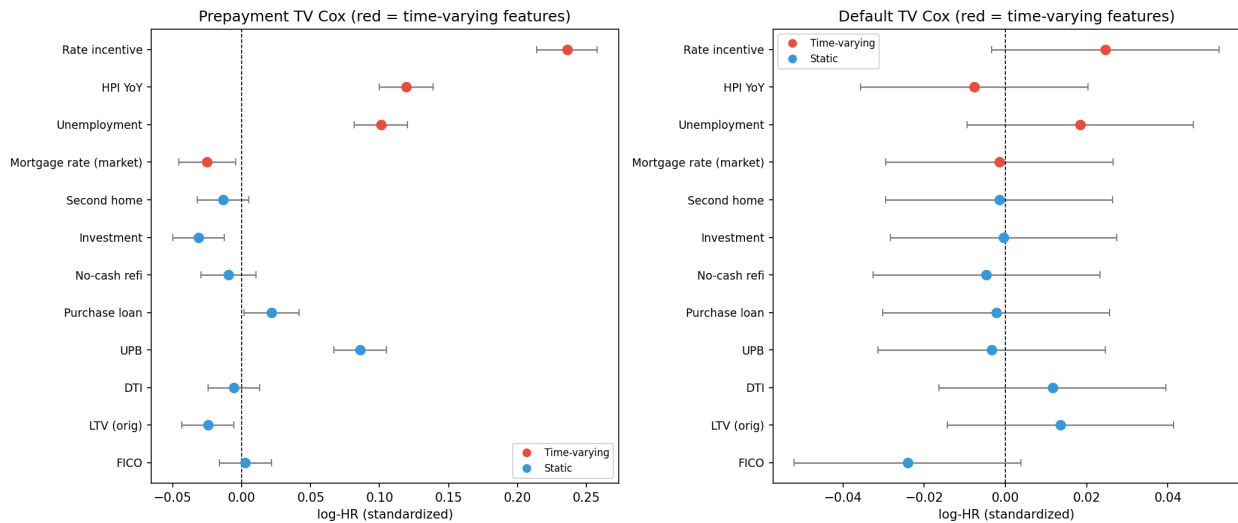


Fig — Eii tv cox forest

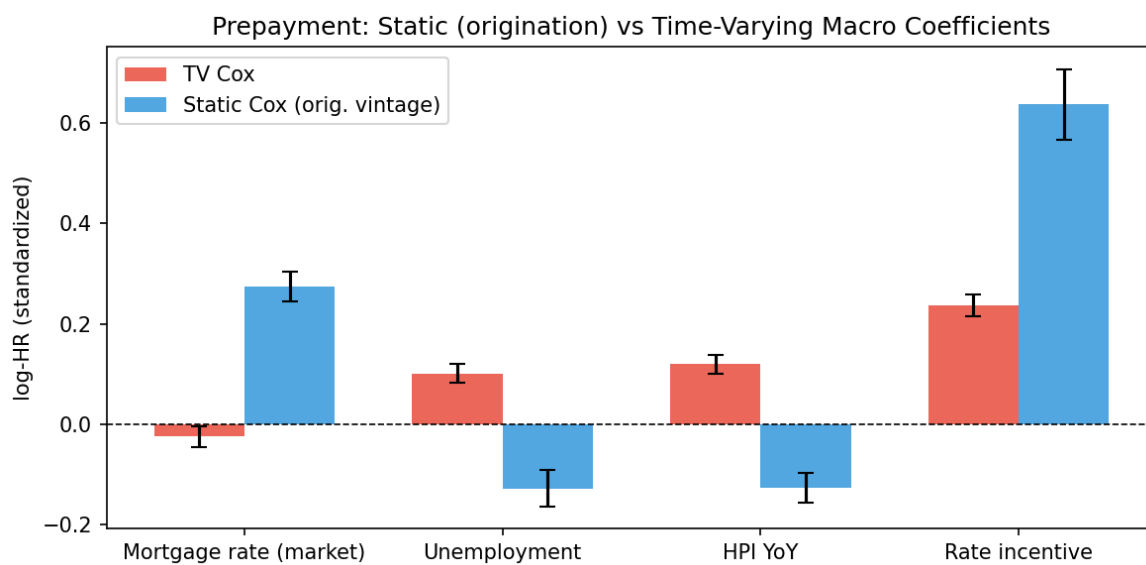


Fig — Eii static vs tv

E(iv) — Scenario Analysis: Interest Rate Shocks

Both Deep Cox and XGBoost are shocked with ± 100 bp and ± 200 bp changes to *mortgage_rate*. Rate incentive is updated consistently: $\text{rate_incentive} = \text{orig_rate} - (\text{mortgage_rate} + \text{shock})$.

Scenario Analysis: Prepayment Sensitivity to Interest Rate Shocks

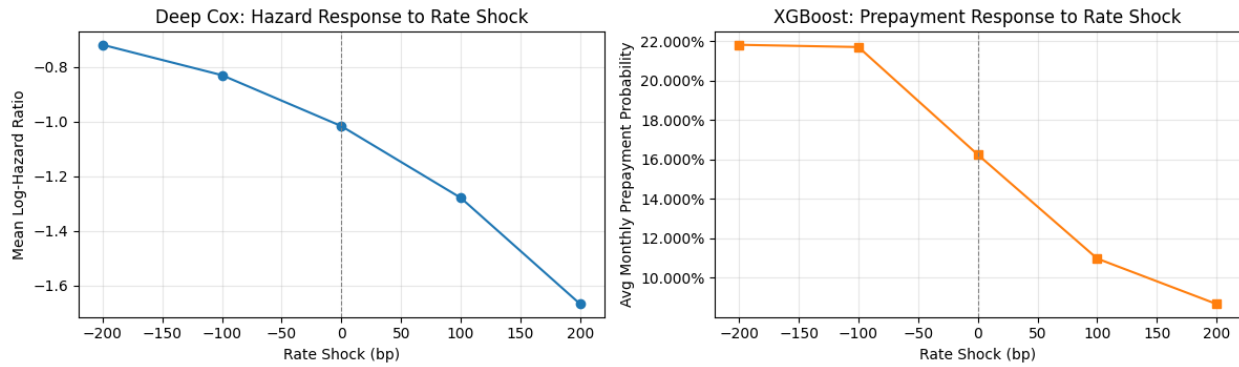


Fig 8. Prepayment sensitivity to interest rate shocks. XGBoost (right) shows clear monotone convex response; Deep Cox (left) shows a smoother log-hazard ratio response.

Interpretation

- XGBoost shows a clear monotone, convex response: rate cuts raise prepayment probability more than equivalent rate rises reduce it.
- Deep Cox responds monotonically with smaller relative changes — the log-hazard is a smoother function of *rate_incentive*.
- Asymmetry (rate cuts > rate rises) reflects the burnout effect: the most rate-sensitive borrowers have already prepaid, dampening upside response.
- Both models agree on direction, confirming robustness of the rate-incentive channel across architectures.

Summary of Part E Extensions

Section	Method	Key Result
E(i)(b) AJ CIF	Aalen-Johansen	10-yr prepay CIF $\approx$ 45%; default $\approx$ 2%
E(i)(c) KM Bias	KM vs AJ comparison	KM over-estimates by +0.5–2.2 pp at 5–20 yr
E(i)(d) Stratified CIF	AJ by FICO / LTV / era	Crisis vintages show 8–12 $\times$ higher default CIF
E(i)(e) Cause-Specific Cox	Two CoxPH models, 12 features	CreditScore: opposite signs across causes
E(i)(f) Fine-Gray	IPCW-weighted Cox	rate_incentive HR larger vs cause-specific
E(ii) TV Cox	Andersen-Gill, 11 features	Current rate_incentive > origination snapshot
E(iv) Scenarios	Deep Cox + XGBoost shock	Convex, asymmetric prepayment response